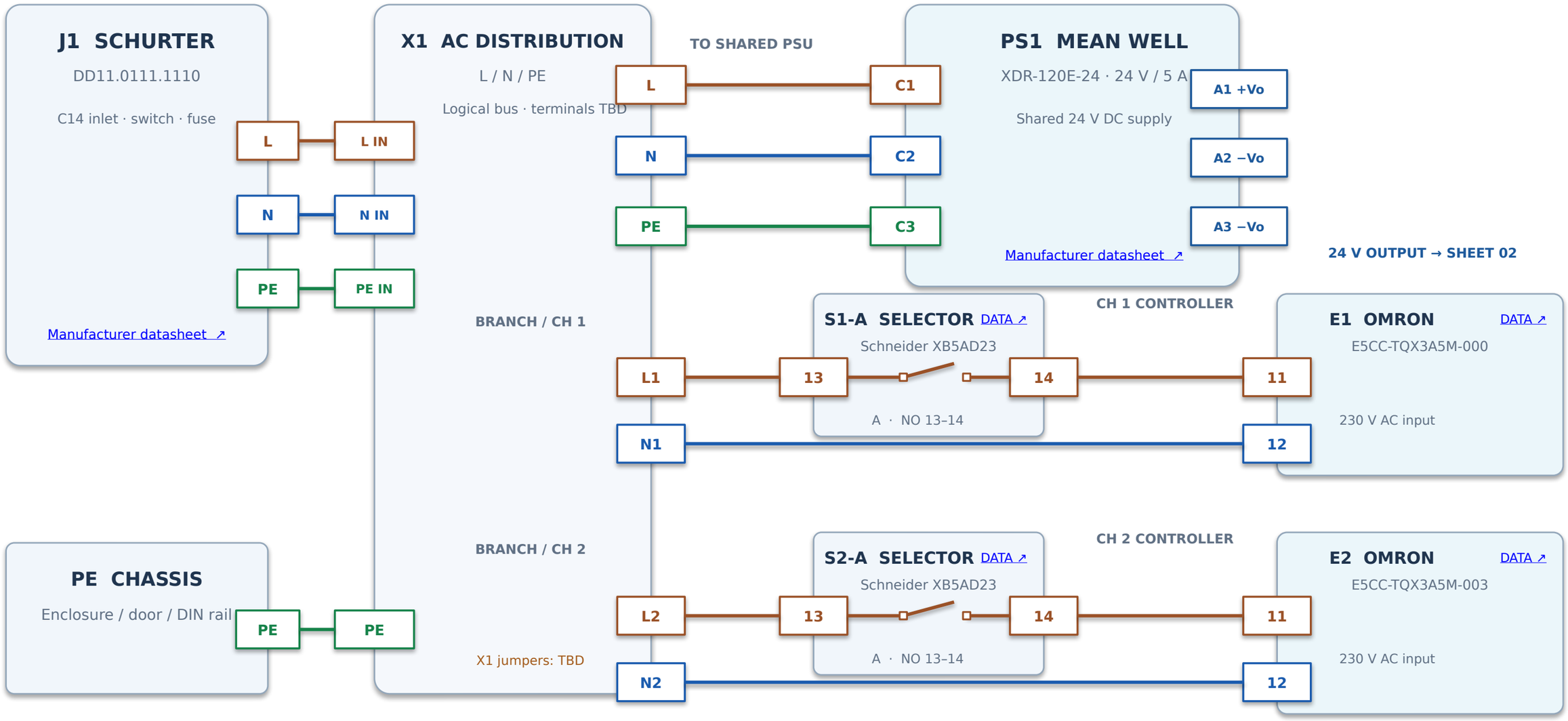
# 01 / MAINS SUPPLY & DISTRIBUTION

L N PE

Two independent Omron channels • Shared 24 V supply • Editable schematic




**DESIGN REVIEW ONLY — Not approved for wiring / energising**

Confirm X1 terminal blocks and jumpers, DD11 output tabs / fuse, PE bonds, and 24 V interlock design.

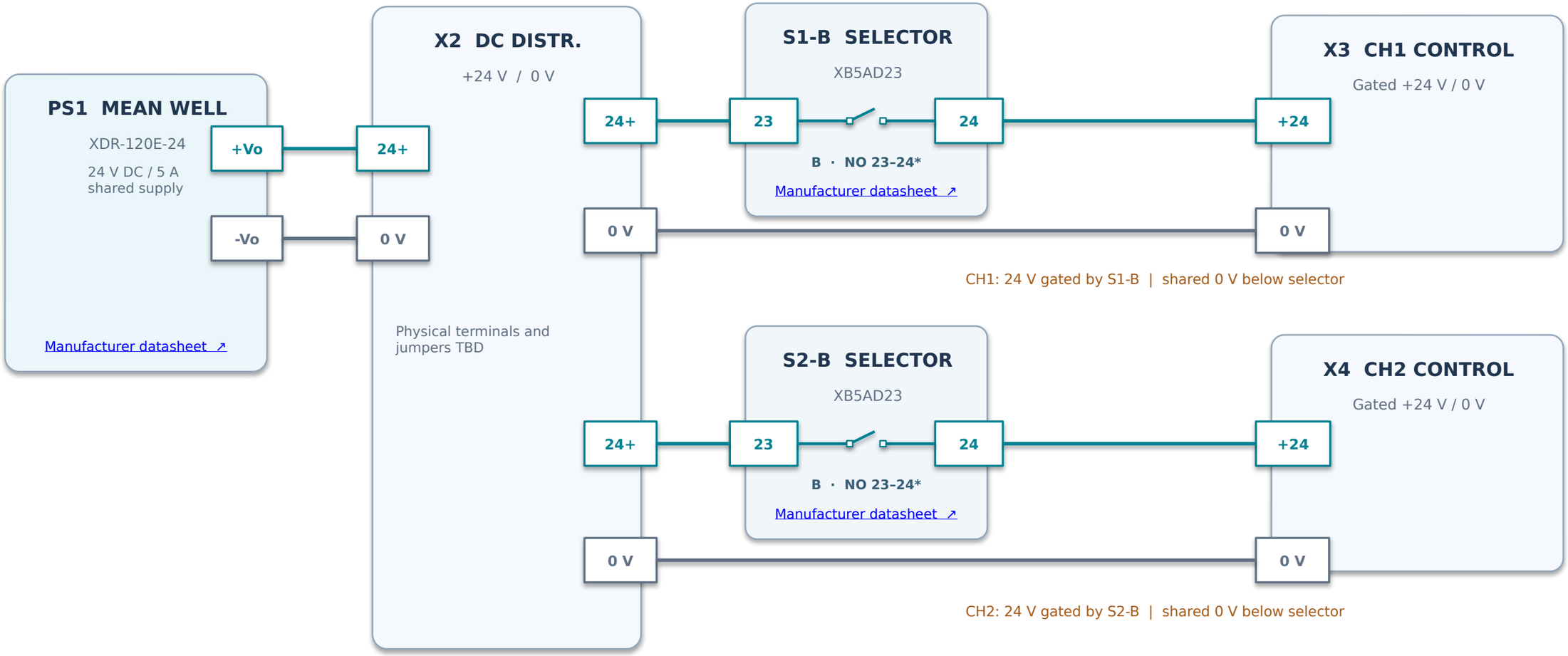
# 02 / SHARED 24 V DISTRIBUTION & CHANNEL INTERLOCKS

+24 V / gated

0 V return

One shared Mean Well supply; S1-B and S2-B independently enable their respective 24 V branches.

## 24 V DC DISTRIBUTION / CHANNEL 1 + CHANNEL 2




Selector terminal references: A = 13–14 (AC) | B = 23–24* (gated +24 V).
* IEC design labels only. Confirm stamps on the actual XB5AD23 contact-block assembly before wiring; X2–X4 distribution hardware remains TBD.

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

# 03 / CHANNEL 1 / MOMENTARY ENABLE & FINDER CONTROL

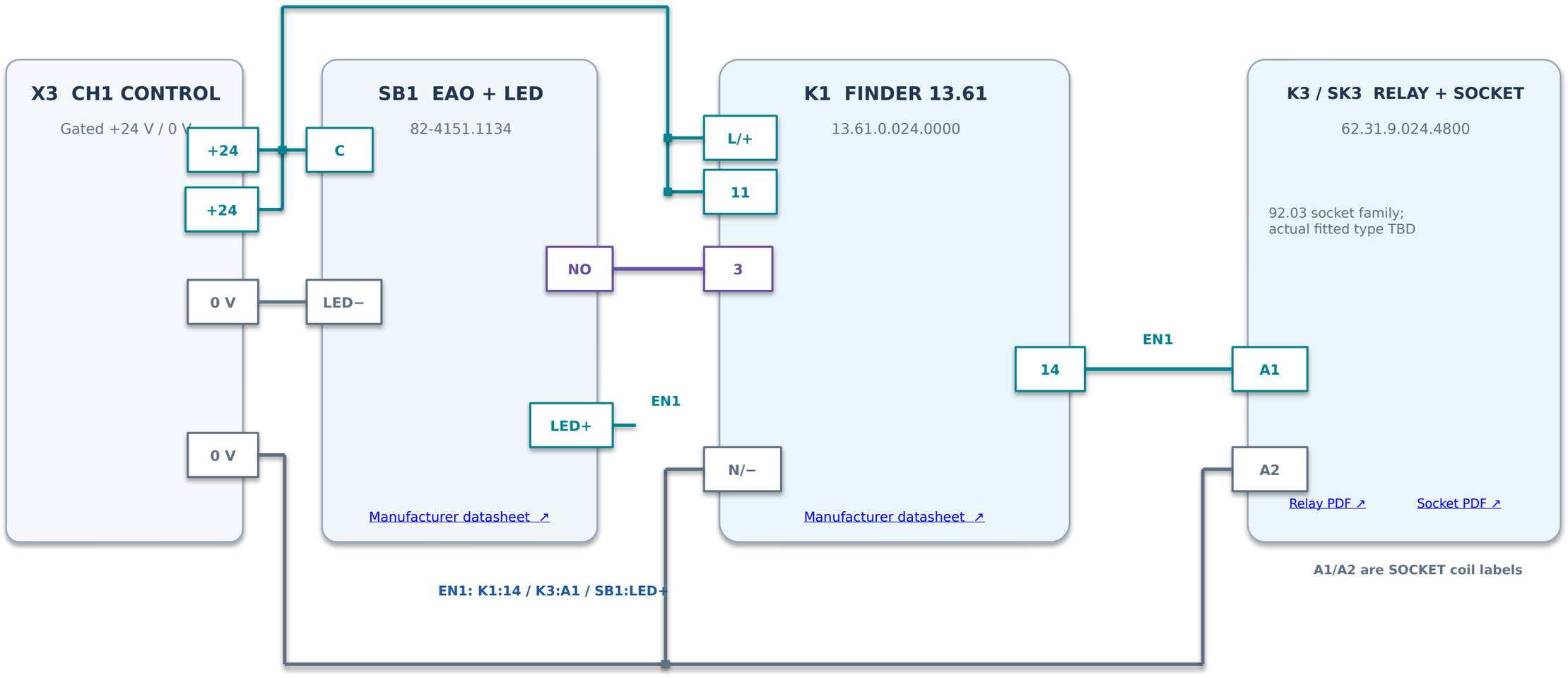
EAO button pulses the Finder 13.61; its output enables the existing Finder 62.31 heater relay (CH1).

Gated +24 V

0 V return

Momentary input

## HEATER ENABLE CIRCUIT / 24 V DC




Matching EN reference indicates ONE electrical net (avoids a wire crossing other circuits).

X3:+24 → button C and K1 L/+,11 | Button NO → K1:3 | EN1 → relay coil A1 and LED+ | 0V → K1:N/−, K3:A2, LED−

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

OFF-on-power-restoration and exact Finder supply/trigger/socket terminal markings remain to be verified.

# 04 / CHANNEL 2 / MOMENTARY ENABLE & FINDER CONTROL

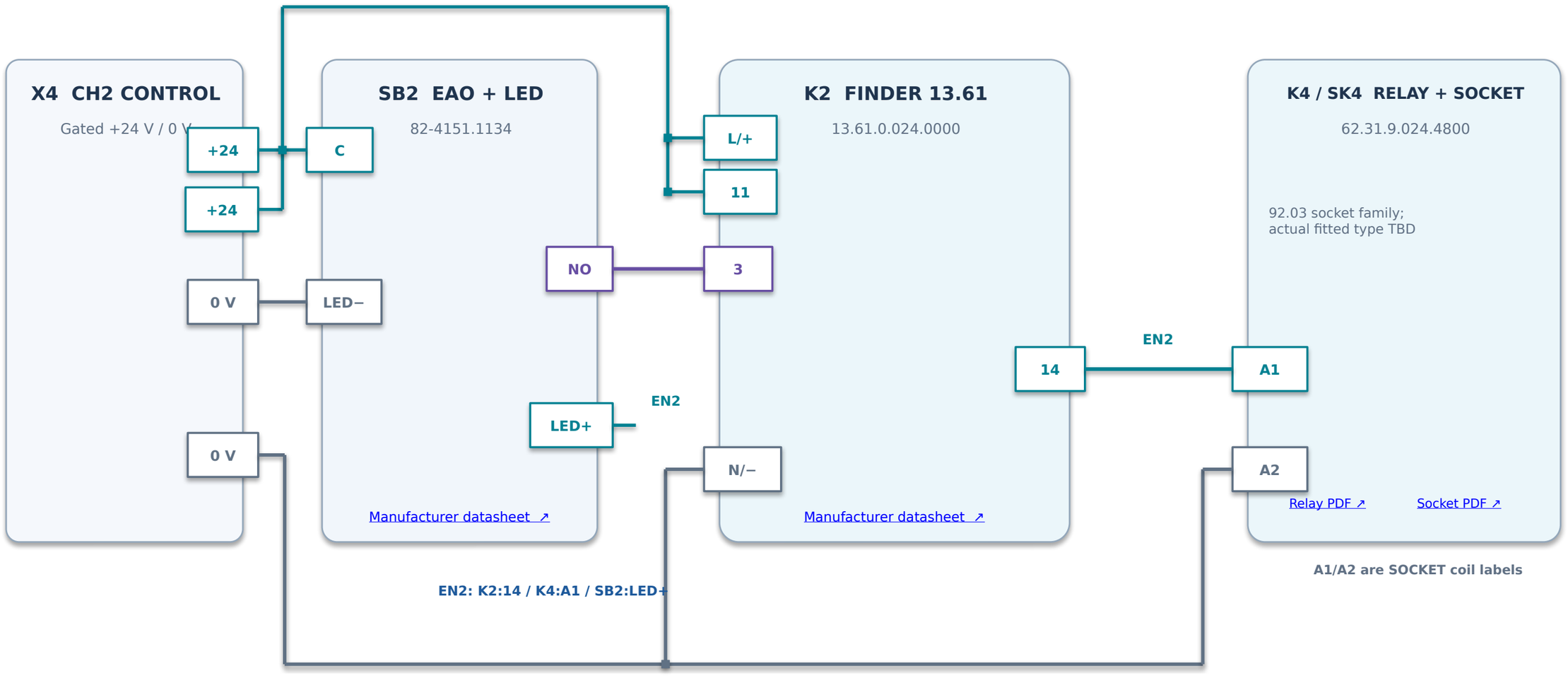
EAO button pulses the Finder 13.61; its output enables the existing Finder 62.31 heater relay (CH2).

Gated +24 V

0 V return

Momentary input

**HEATER ENABLE CIRCUIT / 24 V DC**




Matching EN reference indicates ONE electrical net (avoids a wire crossing other circuits).

X4:+24 → button C and K2 L/+,11 | Button NO → K2:3 | EN2 → relay coil A1 and LED+ | 0V → K2:N/−, K4:A2, LED−

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

OFF-on-power-restoration and exact Finder supply/trigger/socket terminal markings remain to be verified.

# 05 / CHANNEL 1 / SSR DRIVE + HEATER ACTIVITY LED

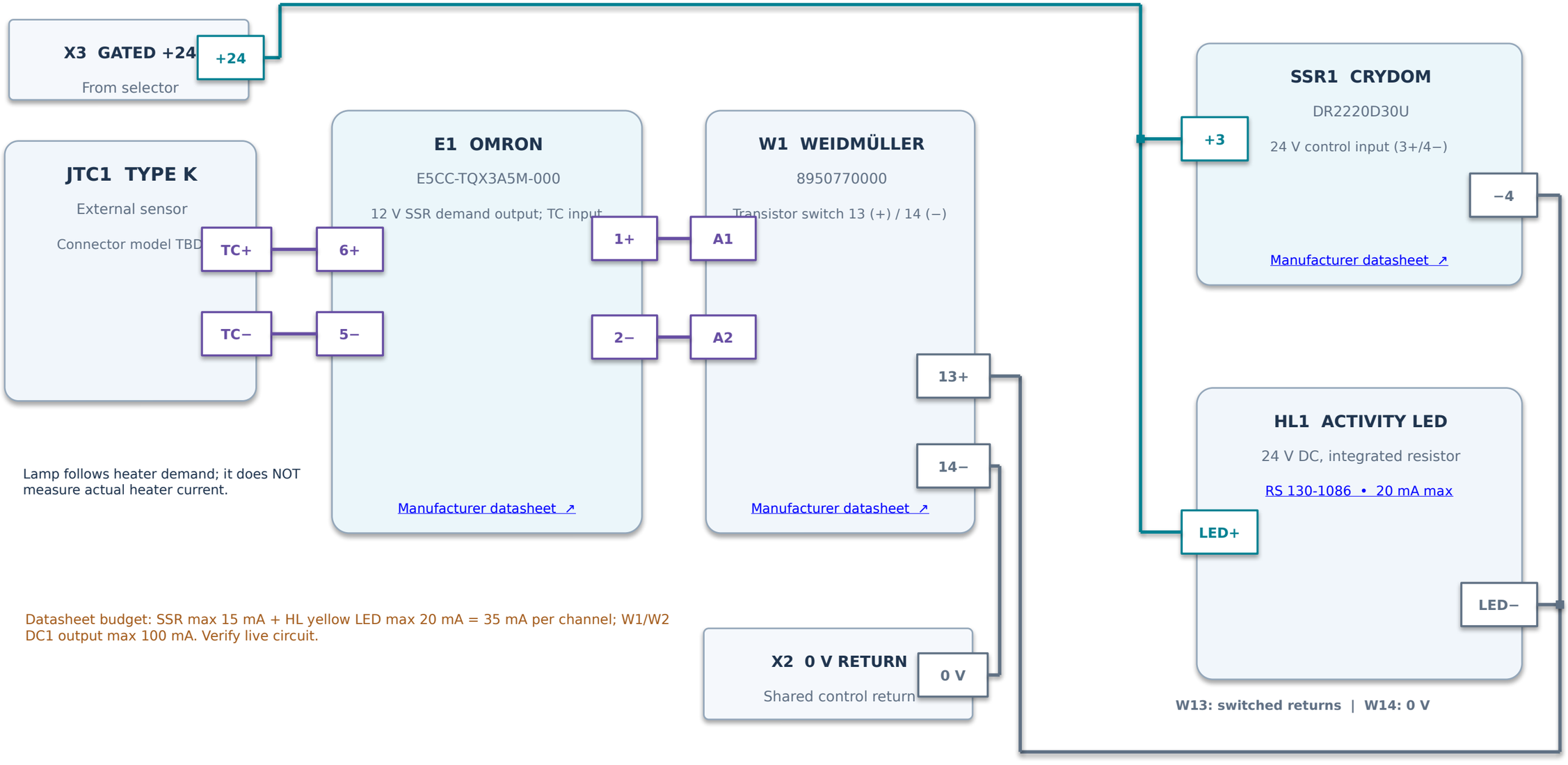
Omron 12 V SSR demand controls an isolated Weidmüller interface switching both Crydom input and activity LED.

TC / 12 V demand

24 V to both loads

Switched return

**THERMOCOUPLE / PULSE INTERFACE / TWO PARALLEL 24 V LOADS**




Lamp follows heater demand; it does NOT measure actual heater current.

Datasheet budget: SSR max 15 mA + HL yellow LED max 20 mA = 35 mA per channel; W1/W2 DC1 output max 100 mA. Verify live circuit.

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

LED is on the 24 V CONTROL circuit, not the external heater: HL1/HL2 = RS 130-1086, yellow, 24 V DC, 20 mA max.

# 06 / CHANNEL 2 / SSR DRIVE + HEATER ACTIVITY LED

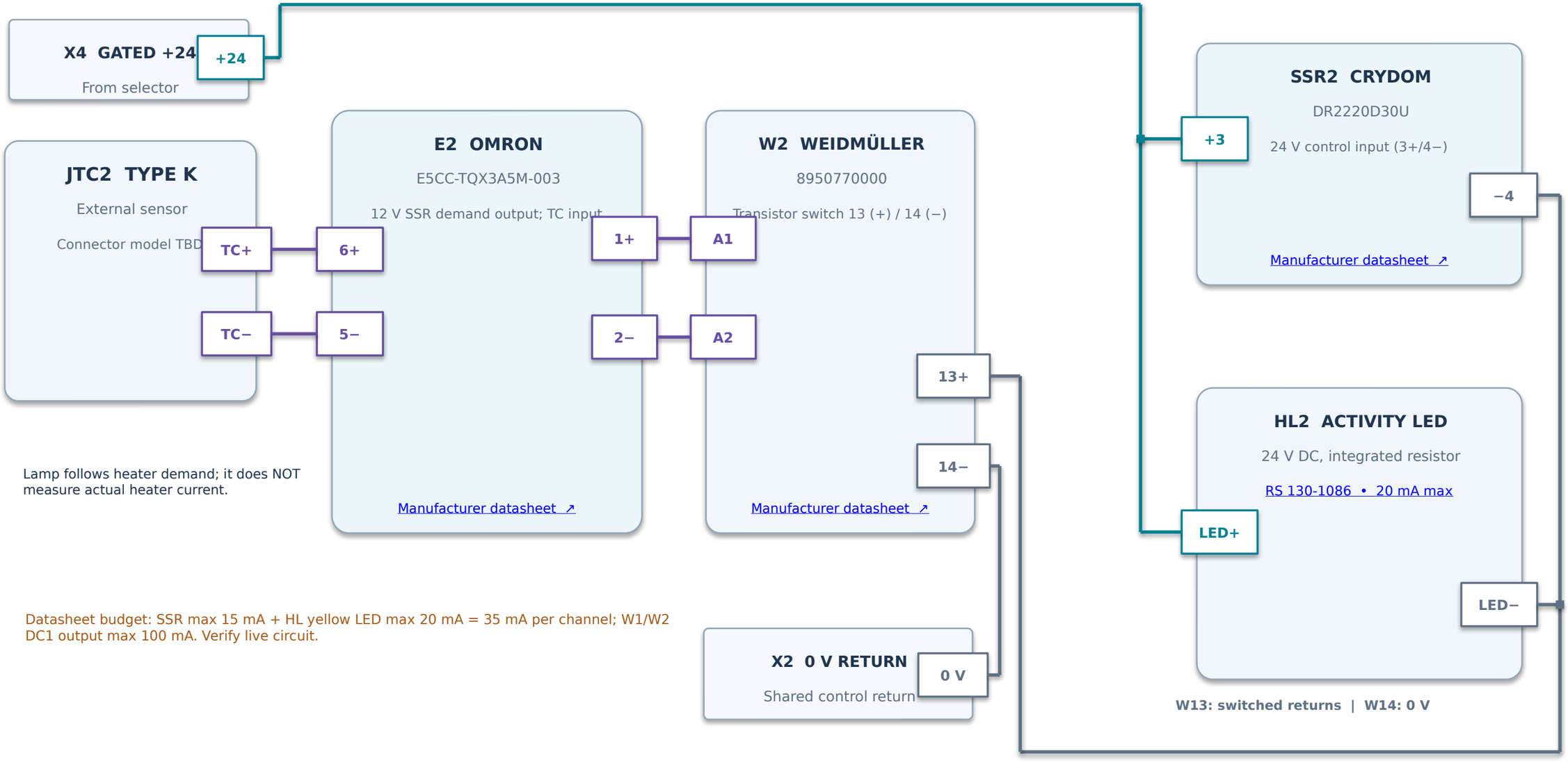
Omron 12 V SSR demand controls an isolated Weidmüller interface switching both Crydom input and activity LED.

TC / 12 V demand

24 V to both loads

Switched return

**THERMOCOUPLE / PULSE INTERFACE / TWO PARALLEL 24 V LOADS**




Lamp follows heater demand; it does NOT measure actual heater current.

Datasheet budget: SSR max 15 mA + HL yellow LED max 20 mA = 35 mA per channel; W1/W2 DC1 output max 100 mA. Verify live circuit.

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

LED is on the 24 V CONTROL circuit, not the external heater: HL1/HL2 = RS 130-1086, yellow, 24 V DC, 20 mA max.

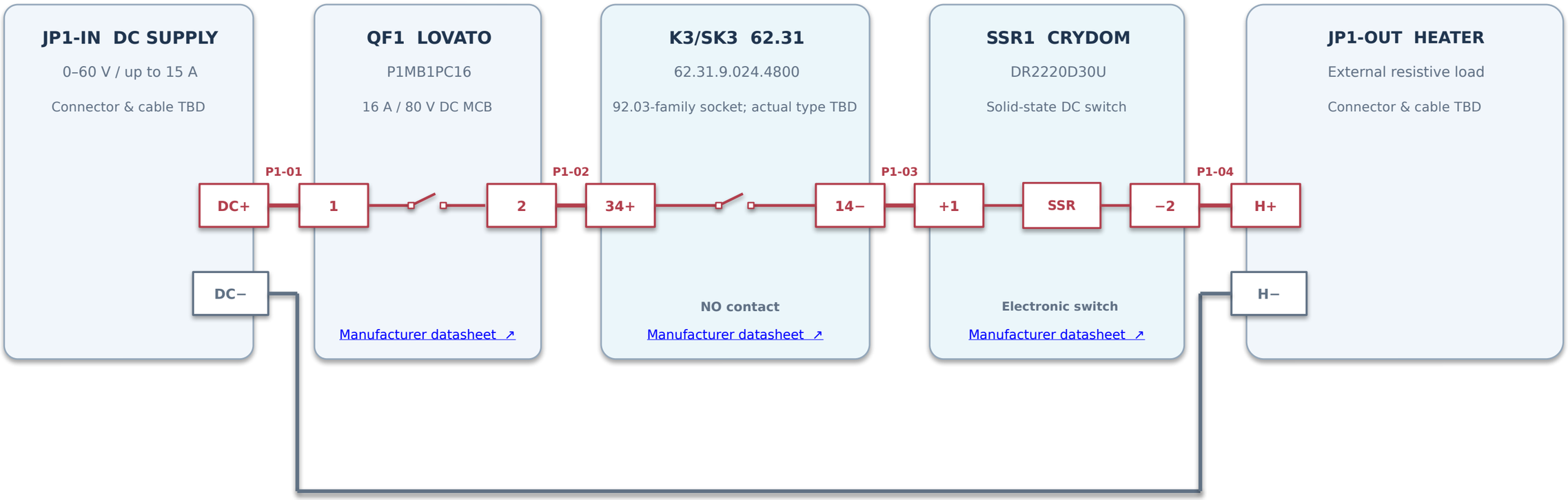
# 07 / CHANNEL 1 / COMPLETE DC HEATER POWER LOOP

External 0–60 V DC supply, breaker, polarity-sensitive Finder DC relay, Crydom SSR and direct return.

Switched positive

Separate DC− return

**INDEPENDENT HEATER SUPPLY / 15 A MAX DESIGN**




**DC NEGATIVE RETURN — SEPARATE FROM SWITCHED POSITIVE**

Finder terminal 14 (−) and SSR load −2 remain in the POSITIVE conductor — neither is the supply 0 V.
Breaker coordination, socket/contact derating and SSR heat dissipation require design approval.

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

Finder contact 34 (+) → 14 (−) and SSR load +1 → −2 are both in the POSITIVE current path.

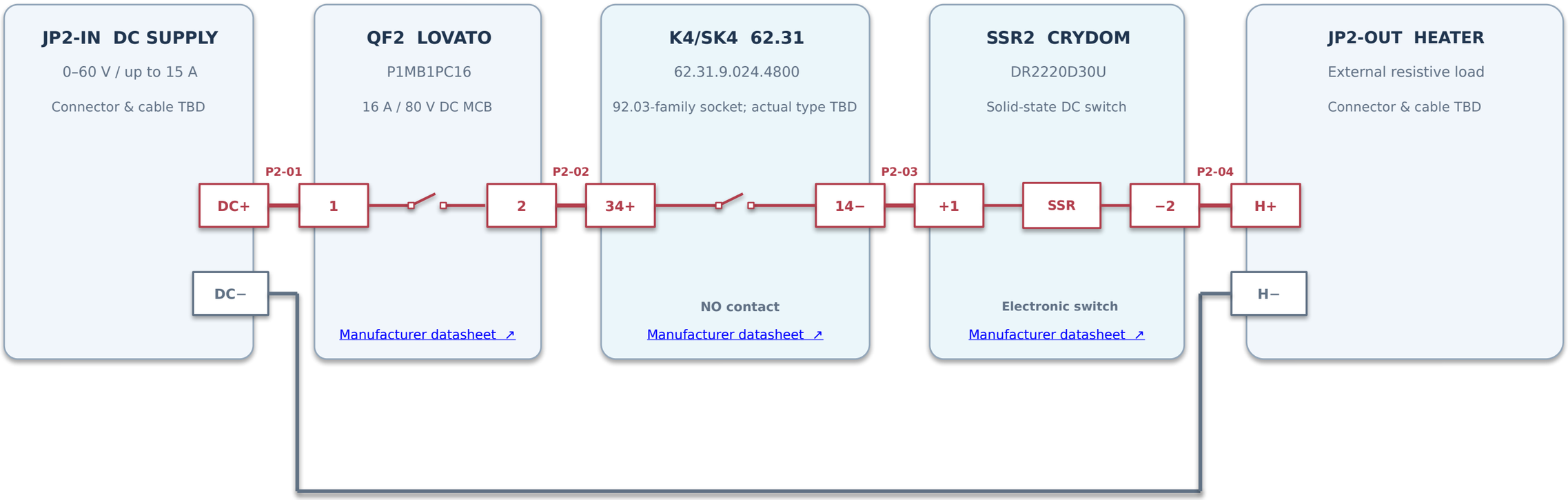
# 08 / CHANNEL 2 / COMPLETE DC HEATER POWER LOOP

External 0–60 V DC supply, breaker, polarity-sensitive Finder DC relay, Crydom SSR and direct return.

Switched positive

Separate DC− return

**INDEPENDENT HEATER SUPPLY / 15 A MAX DESIGN**




**DC NEGATIVE RETURN — SEPARATE FROM SWITCHED POSITIVE**

Finder terminal 14 (−) and SSR load −2 remain in the POSITIVE conductor — neither is the supply 0 V.
Breaker coordination, socket/contact derating and SSR heat dissipation require design approval.

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

Finder contact 34 (+) → 14 (−) and SSR load +1 → −2 are both in the POSITIVE current path.

# 09 / CONNECTION SCHEDULE / MAINS + 24 V DISTRIBUTION

Functional point-to-point net destinations for the component-port diagrams (not a physical ferrule/cable schedule).

## TERMINAL-LEVEL REFERENCE / MAINS AND 24 V

| | | | | |
|---|---|---|---|---|
| | **AC01** | J1 switched fused L | → | X1:L bus |
| | **AC02** | J1 switched N | → | X1:N bus |
| | **AC03** | J1:PE | → | X1:PE + chassis bonds |
| | **AC04** | X1:L / N / PE | → | PS1:C1 / C2 / C3 |
| | **AC05** | X1:L → S1-A:13/14 | → | E1:11 (L) |
| | **AC06** | X1:N bus | → | E1:12 (N) |
| | **AC07** | X1:L → S2-A:13/14 | → | E2:11 (L) |
| | **AC08** | X1:N bus | → | E2:12 (N) |
| | **DC01** | PS1:A1 +Vo | → | X2:+24 V shared |
| | **DC02** | PS1:A2/A3 −Vo | → | X2:0 V shared |
| | **DC03** | X2:+24 → S1-B:23/24* | → | X3:CH1 gated +24 |
| | **DC04** | X2:+24 → S2-B:23/24* | → | X4:CH2 gated +24 |
| | **DC05** | X2:0 V shared | → | X3 / X4 returns |

Before wiring: confirm selector B contact-block stamps (*23/24 IEC proposed), J1 tabs, PE, X1–X4 blocks/jumpers and mains/24 V fuses.

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

Any X1–X4 numbers are logical net references until actual terminal blocks/jumpers are documented.

# 10 / CONNECTION SCHEDULE / CONTROLS + HEATER

Both channels follow equivalent functional nets; this summary complements the illustrated component-port sheets.

## TERMINAL-LEVEL REFERENCE / CHANNELS 1 AND 2

| | |
|---|---|
| **ENABLE / EACH CH** | X3/4:+24 → Finder 13.61 power & 11; button C. X2:0V → Finder supply return, SK3/4:A2 and EAO LED−. |
| **MOMENTARY INPUT** | EAO SB1/2:NO → Finder K1/2:3 impulse; EAO C uses channel-gated +24 V. |
| **FINDER OUTPUT** | K1/2:14 (NO) → Finder 62.31 SOCKET SK3/4:A1 and EAO LED+; 11 COM fed from gated +24 V. |
| **THERMOCOUPLE** | Type K sensor (+) → E1/E2:6; sensor (−) → E1/E2:5. |
| **OMRON / WEIDMÜLLER** | E1/E2 SSR pulse terminals 1 (+),2 (−) → W1/W2 A1 (+),A2 (−). |
| **WEID / SSR DRIVE** | Gated +24 → SSR+3 AND activity HL+; SSR−4 AND activity HL− → W:13; W:14 → 0V. |
| **DC POSITIVE** | Supply DC+ → Lovato 1→2 → Finder 62.31 contact 34 (+)→14 (−) → SSR +1→−2 → heater+. |
| **DC RETURN** | Heater− → matching external DC supply−; Finder 14 and SSR −2 are NOT DC−. |
| **DC COIL** | Finder 62.31 uses socket SK3/4: coil A1/A2, contact 34(+)/14(−). Actual 92.03 suffix and suppressor TBD. |

These are net-level circuit proposals, NOT numbered physical wires. Final wire IDs, conductor sizes, terminations and current ratings remain to be assigned.

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

The Finder 13.61 power-up OFF state, switch actual terminals and DC protection must still be verified.

# 11 / MANUFACTURER DOCUMENTS & RELEASE CHECKS

Clickable component documents are embedded inside each named component block throughout the PowerPoint.

## COMPONENT REGISTER / TECHNICAL REVIEW

| | | | |
|---|---|---|---|
| **Schurter DD11.0111.1110** | Datasheet ↗ | **Mean Well XDR-120E-24** | Datasheet ↗ |
| **Schneider XB5AD23 ×2** | Datasheet ↗ | **Omron E5CC-TQX3A5M-000 / -003** | Datasheet ↗ |
| **EAO 82-4151.1134 ×2** | Datasheet ↗ | **Finder 13.61.0.024.0000 ×2** | Datasheet ↗ |
| **Finder 62.31.9.024.4800 ×2** | Datasheet ↗ | **Weidmüller 8950770000 ×2** | Datasheet ↗ |
| **Crydom DR2220D30U ×2** | Datasheet ↗ | **Lovato P1MB1PC16 ×2** | Datasheet ↗ |

HL1/HL2 specified: RS PRO 130-1086 yellow LED, 24 V DC, max 20 mA, IP67; 14 mm mounting hole.

## REMAINING DESIGN-RELEASE BLOCKERS

**01** Finder 13.61 must remain OFF after loss and restoration of 24 V; prove minimum interruption and button held states.

**02** Confirm X1–X4 blocks, Schneider S1/S2 B-contact terminal stamps (23/24* IEC intended), J1 tabs and Finder 62.31 sockets.

**03** Verify 15 A DC continuity ratings, breaker/source fault coordination, cable size and connector ratings.

**04** Validate SSR heat sink, W1/W2 100 mA output (SSR+LED 35 mA max calculated), PE and emergency disconnection.

**05** Independent safety design review and pre-energisation electrical/function tests by a suitably competent person.

Finder 92.03-series socket datasheet (actual fitted socket still unverified) ↗

**DESIGN REVIEW ONLY | Terminal references are proposed where physical hardware is unidentified. Not approved for wiring / energising.**

Manufacturer sources are reference points, not proof that the integrated design has passed electrical safety review.